

EventSure

Your Event, Protected

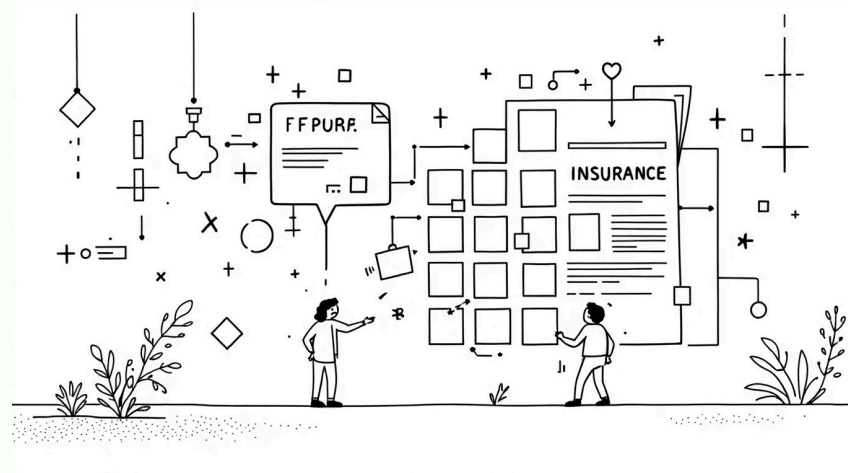
A Full-Stack Event Insurance Platform

Capstone Project Presentation By **Lakshmi Srujana G**



The Problem in the Insurance Space

Traditional event insurance is slow and frustrating.



Complex Manual Process

Manual processing of insurance requests, slow claim settlement, and lack of transparency between customers and agents.

Lack of Understanding

Complexity in understanding policy terms.

Fragmented Agency Workflows

Agents and claims officers struggle with disconnected systems, making it difficult to track leads, manage policies, and handle disputes efficiently.

The Solution: EventSure

A unified ecosystem for instant coverage and tracking.



Centralized Platform

Connects Customers, Agents, Admins, and Claims Officers in one unified ecosystem, streamlining communication and workflows.



Tailored Coverage

Dynamically matches policy templates to specific event risks (alcohol, outdoors, VIPs), ensuring precise and relevant coverage for every event.



Fast processing

Reduces policy processing time by up to 60%.

Tech stack & System Architecture

- **Frontend:** Angular 20+ (Standalone Components), TailwindCSS4 (Styling), RxJS (State Management).
- **Backend:** .NET 8 Web API, Entity Framework Core.
- **Architecture:** Clean Architecture (Domain-Driven Design), Repository Pattern.
- **Database:** SQL Server (Relational data for users, policies, and claims).
- **Security:** JWT Authentication, Role-Based Access Control (RBAC), xUnit & Vitest and MFA challenges.

Layers:

- a. **UI Layer:** Responsive Angular components.
 - b. **Service Layer:** Business logic and external API integrations (Auth, Payment processing).
 - c. **Data Layer:** SQL Server persistence via EF Core.
- **Interceptors:** Global error handling and JWT injection.

System Roles & Portals

Secure, Role-Based Access Control (RBAC)



Customer Portal

Create profiles, send quotes, receive policy suggestions, process payments securely, and file claims with ease.



Admin Dashboard

Oversee system metrics, create baseline policy products, and efficiently assign agents to new leads.



Agent Workplace

Dedicated CRM to manage assigned requests, send customized policy suggestions, and track personal commissions.



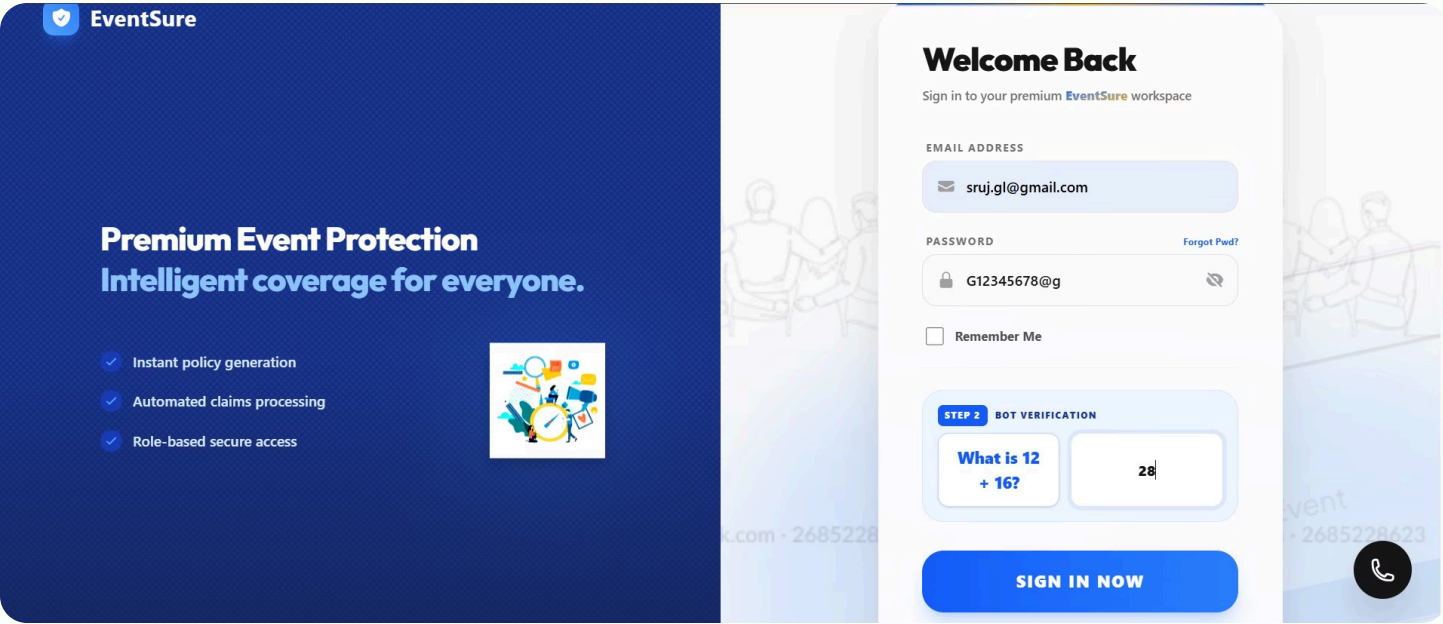
Claims Officer Portal

Specialized workflow designed to investigate, accept, or reject filed claims safely and promptly.

Feature Highlight

The Customer portal

1. User Onboarding & Secure Access.



2. Event Insurance Request Submission: Customers complete a dynamic event insurance request form.

Create Insurance Request

Protect your event with tailored coverage. Provide the details below to receive a professional insurance proposal.

01 Event Information

CORE DETAILS OF YOUR UPCOMING EVENT

EVENT TYPE

Select Category

DATE AND TIME

DD-MM-YYYY --:--

EVENT DURATION (HOURS)

1

VENUE / LOCATION

AUTO LOCATE

1. Policy Suggestions & Review: Here agents analyze the event details and suggest suitable insurance policies
2. Policy Activation & Payment.

My Active Policies

Your approved and active insurance coverage.

Policy Portfolio						
POLICY #	NAME	COVERAGE	VALID UNTIL	PAYMENT PLAN	PAID PROGRESS	ACTIONS
EVT-20260311-9	Conf Standard	\$200,000.00	Mar 11, 2027	Monthly Total: \$4,600.00	PAID: 2 / 12 17% Due: May 11, 2026	Pay Due (\$383.33)

3. Claim Management (If Required).

My Claims

File and track insurance claims on your active policies.

File a New Claim

Select Active Policy

EVT-20260311-9 — Coverage: \$200,000.00

Claim Amount Requested

50000

Max claim: \$200,000.00

Claim Reason

due to event cancellation

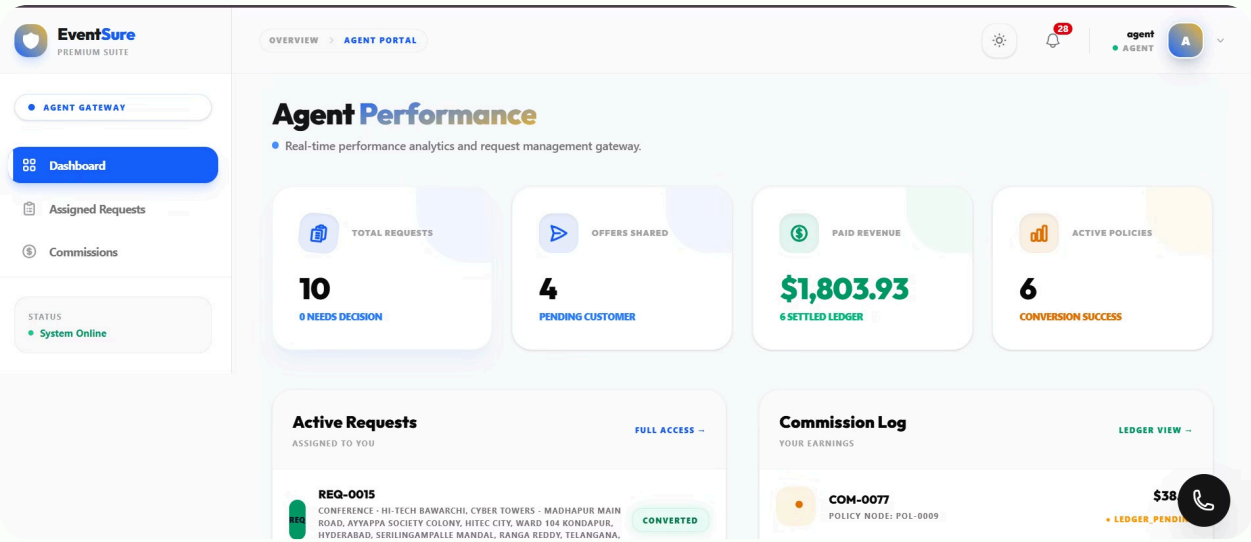
Submit Claim

4. Self-Service Support & Help Center.

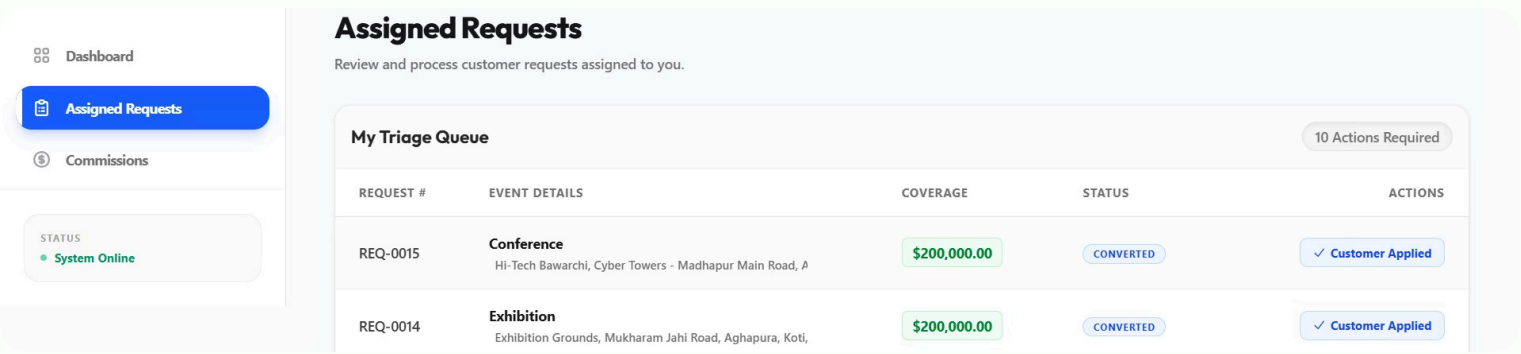
Login → Submit Request → Agent Suggests Policy → Customer Selects Policy → Admin Approves → Payment → Active Policy → Claim (Optional)

The Agent portal

1. Secure Login & Dashboard Access.

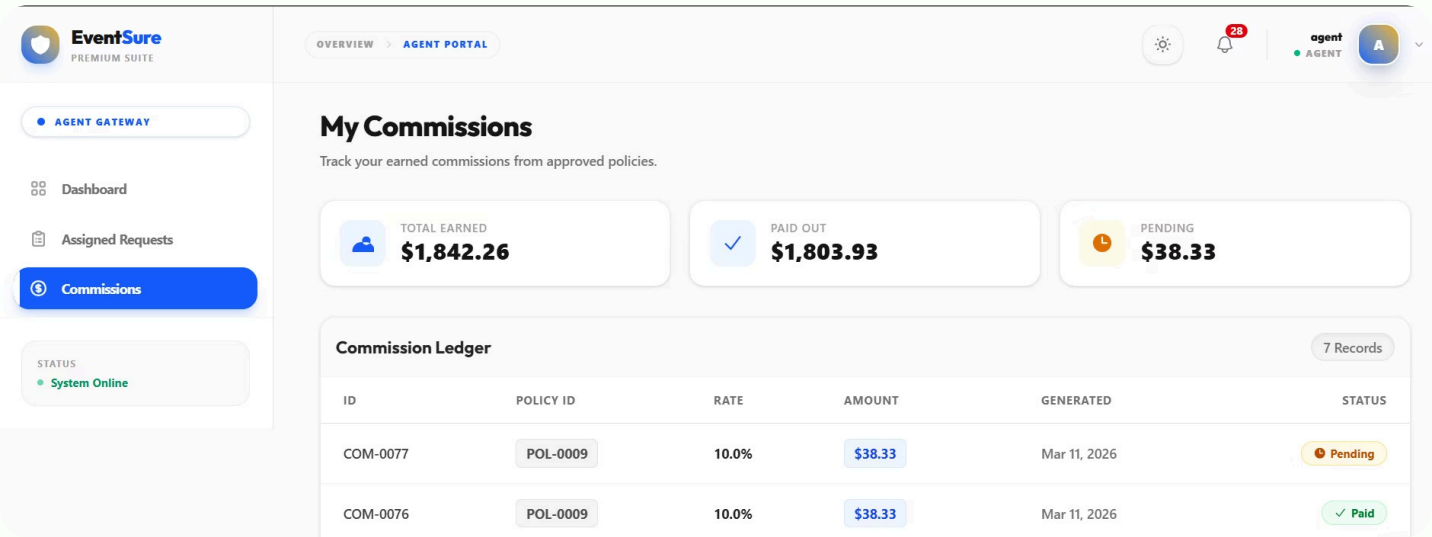


2. Assigned Request Management: Reviews customer event insurance requests assigned by the admin.



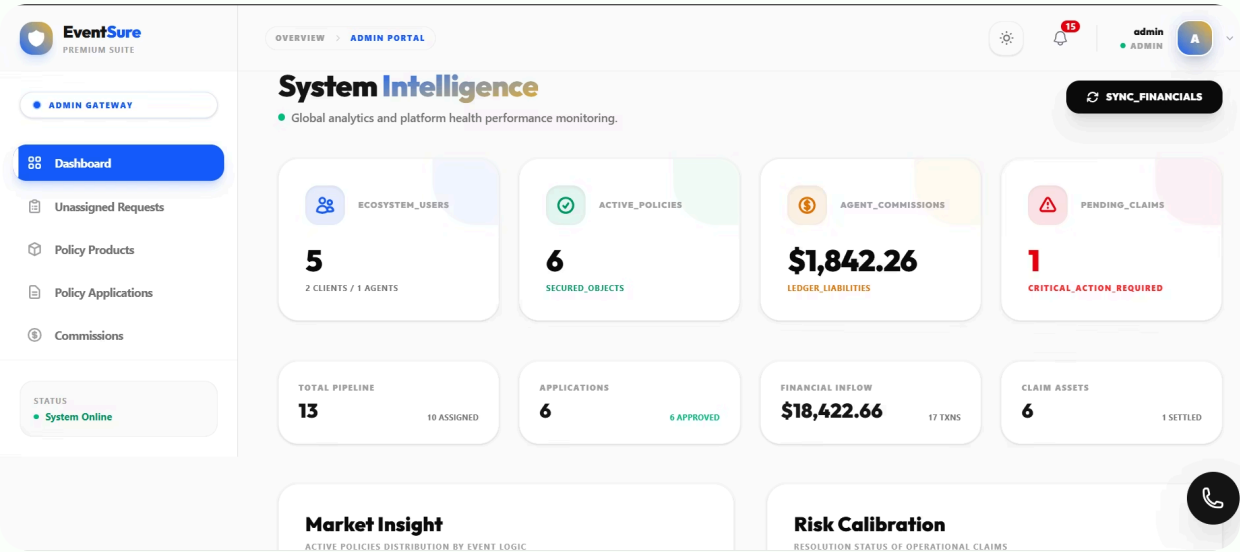
3. Policy Analysis & Suggestion: Evaluate suitable policy products and send tailored policy suggestions to customers.

4.Commission Tracking.

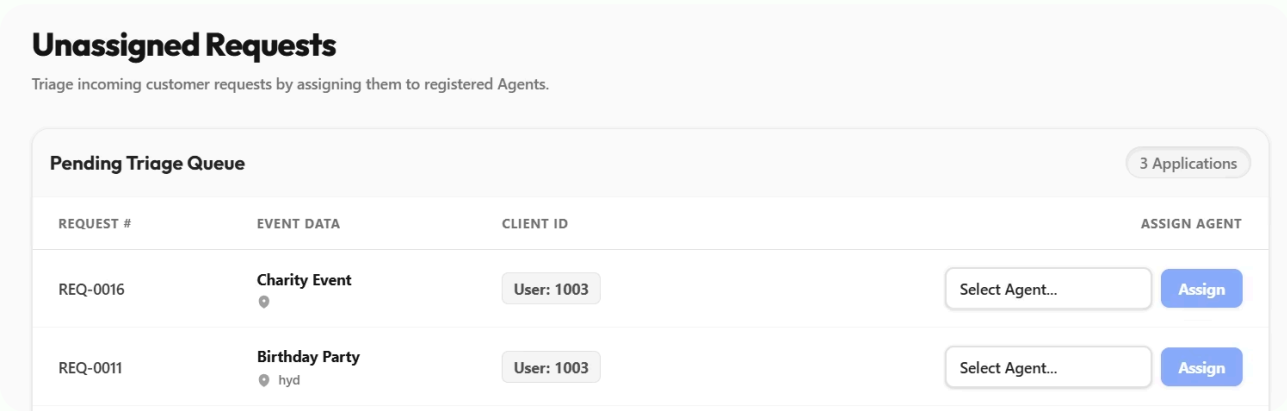


The Admin portal

1. Secure Administrative Access.



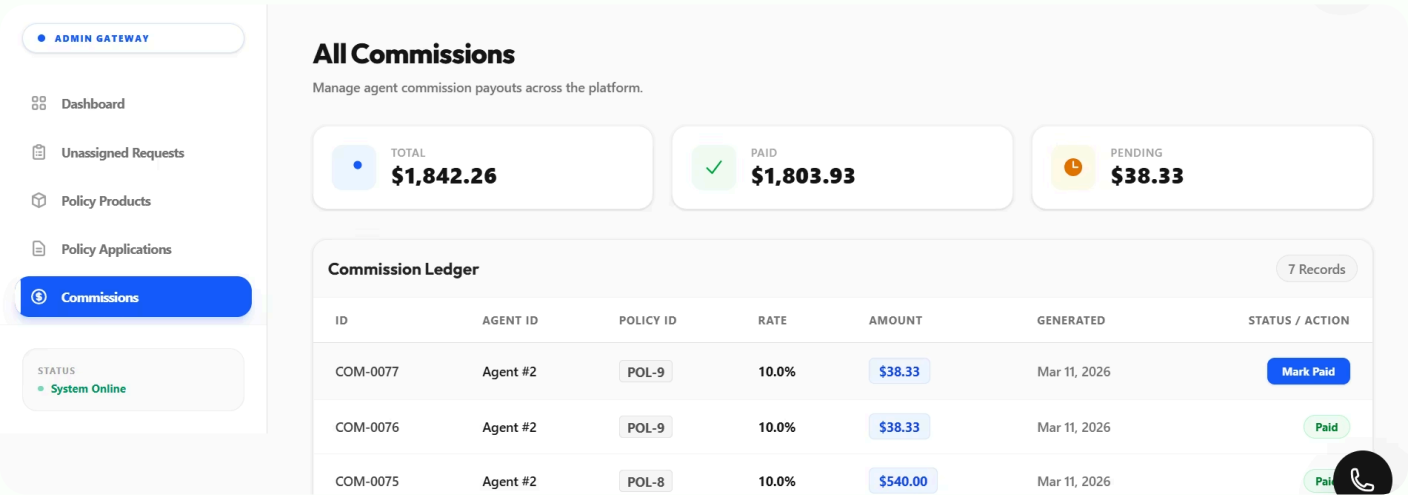
2. Request Monitoring & Agent Assignment



3. Policy Product Management: Create, update, activate, or deactivate insurance policy products.

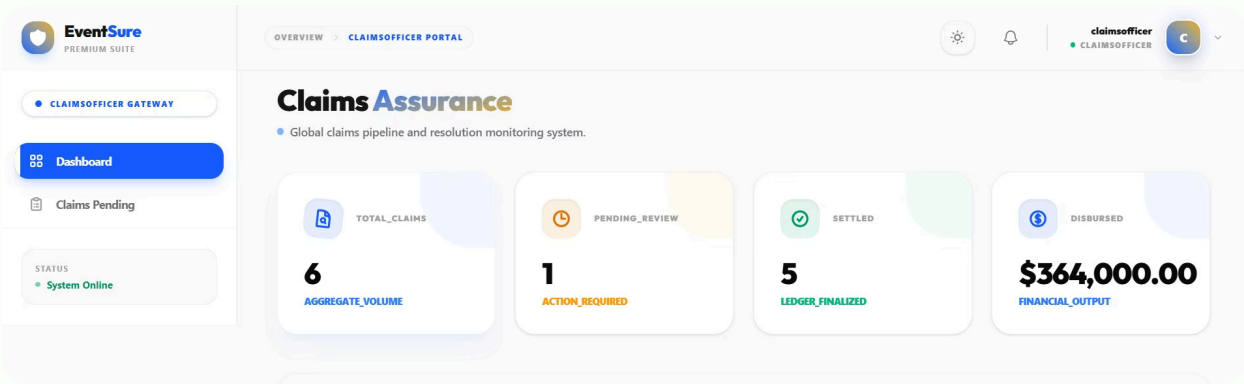
4. Policy Application Approval.

5. Commission & System Reporting: Monitor agent commissions, platform activity, and overall system reports.



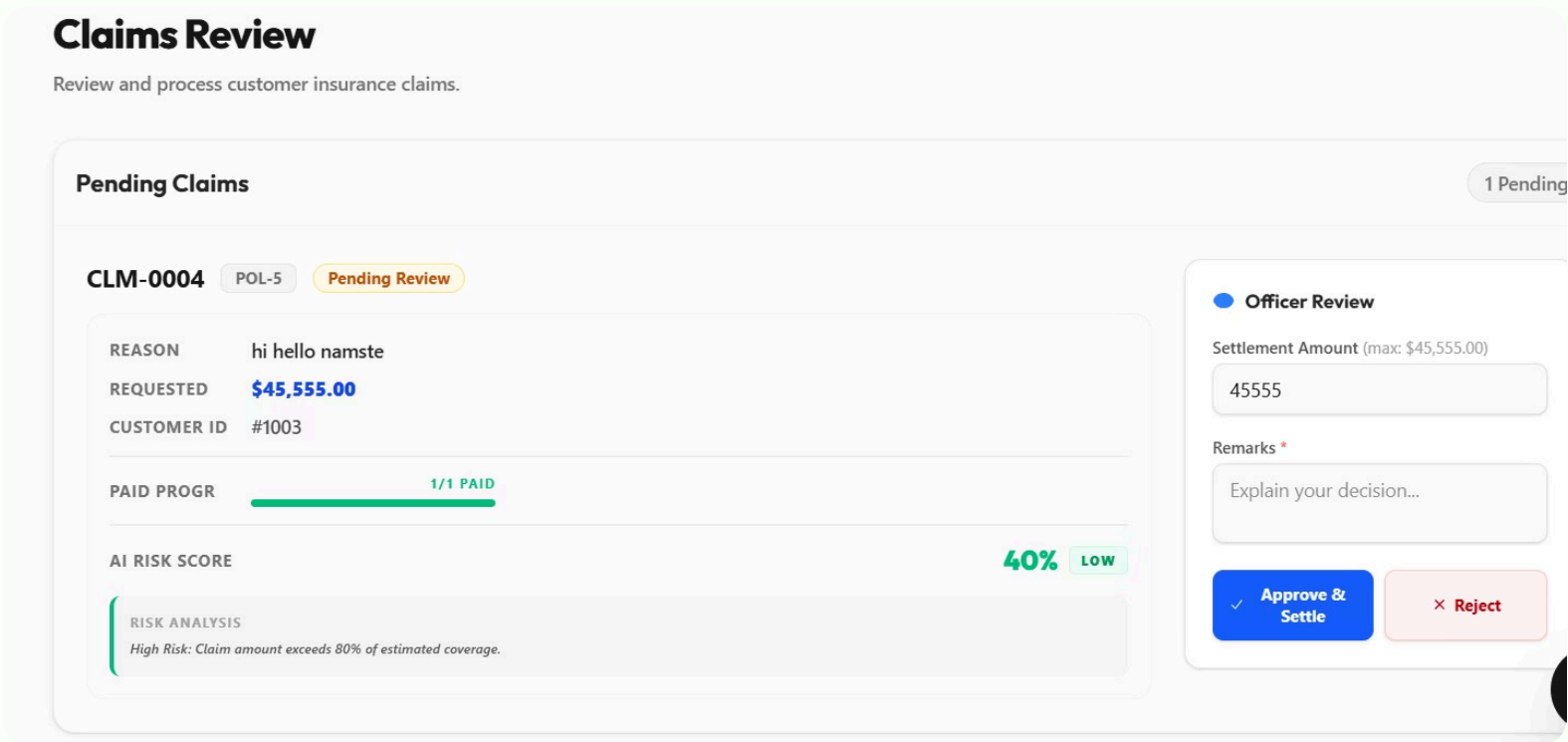
The Claims Officer portal

1. Secure Claims Officer Login



2.Claim Review & Investigation: Reviews customer-submitted claims.

3.Claim Verification Process: Risk scoring based AI verifies eligibility and assess potential settlement amounts.



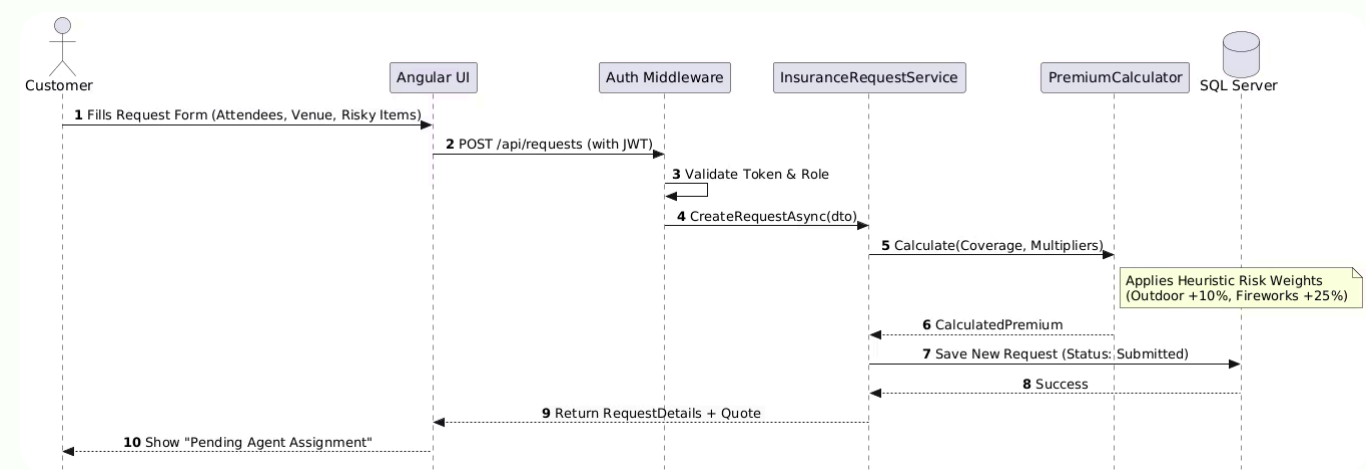
4.Claim Decision & Settlement: Approves or rejects the claim.

5.Customer Notification & Status Update.

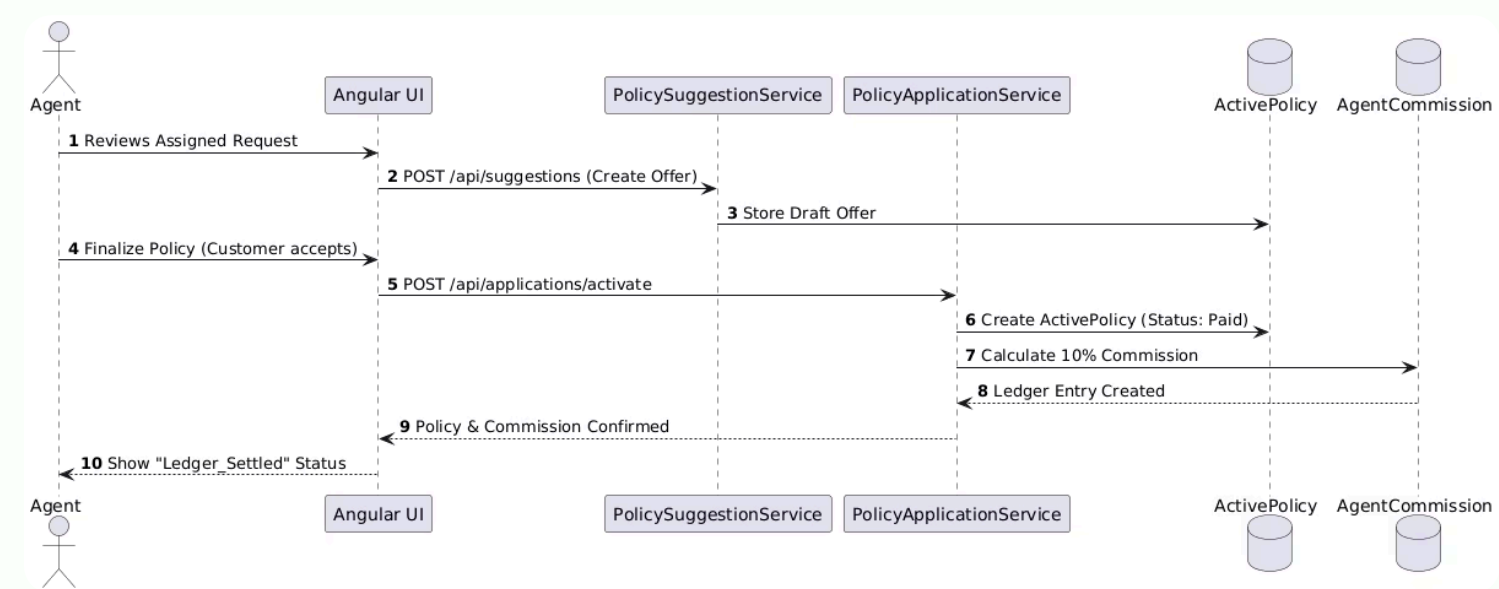
Project Flow: Sequence

EventSure transforms the customer experience into a clear, intuitive journey, guiding them from initial interest to comprehensive protection. From initial inquiry to final claim resolution, EventSure provides a seamless and supportive experience for every customer:

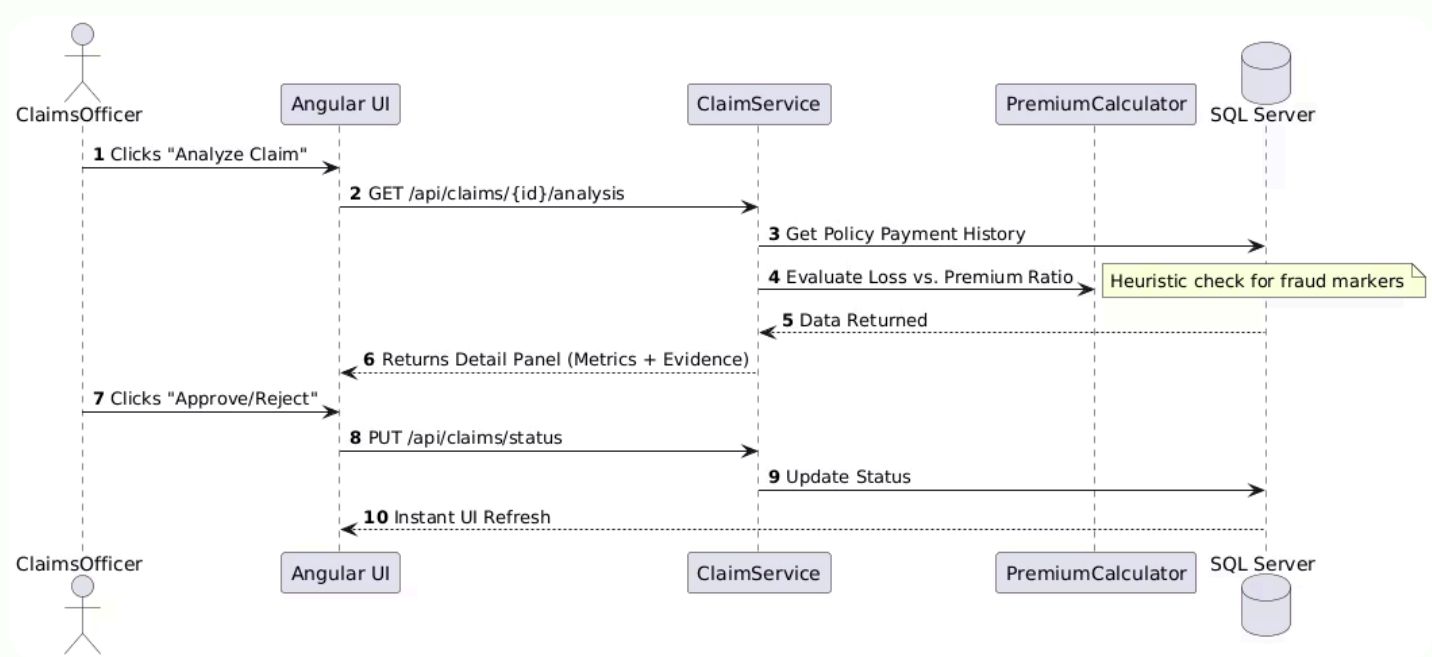
1. Insurance Request & Quote Flow (Customer + Heuristic Engine)



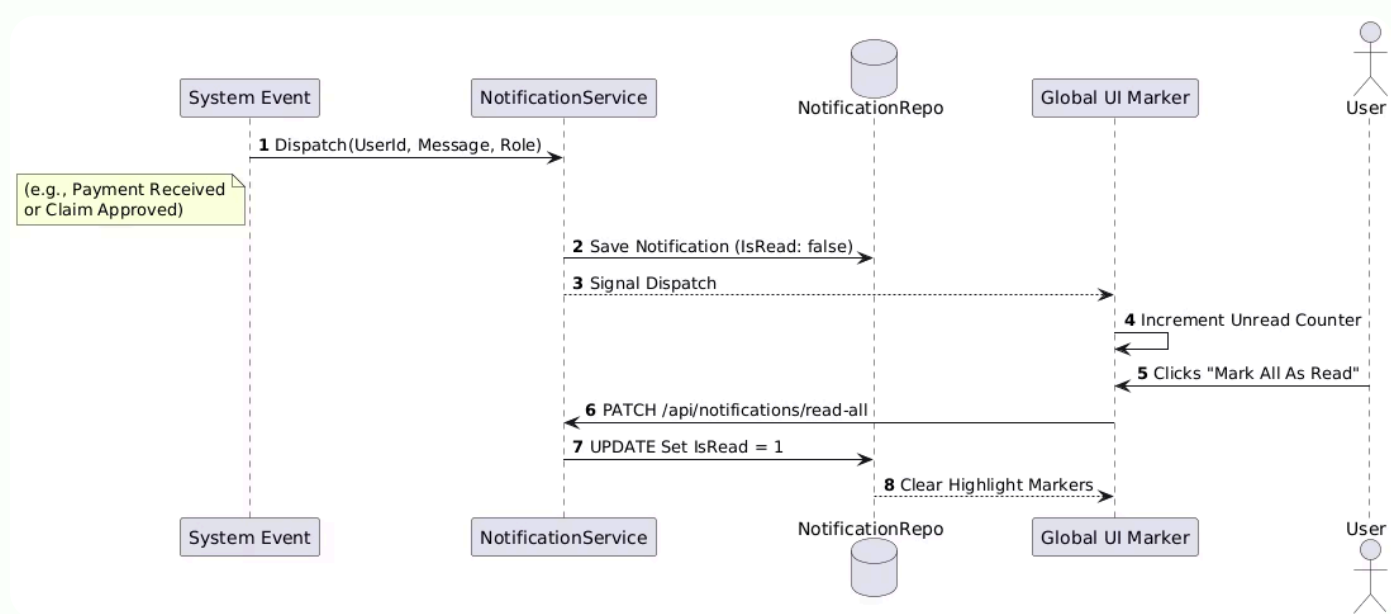
2. Agent Workflow (Policy Generation & Commission)



3. Claims Handling Flow (Audit & Fraud Analysis)



4. Real-time Notification Dispatch



Key Features & Capabilities

Built for Scale and Usability

- **Multi-Role Ecosystem:** Four fully distinct portals (Customer, Agent, Admin, Claims Officer) sharing a unified notification and authentication bridge
- **Dynamic Ad Engine:** A hero section featuring an auto-rotating ad carousel with custom-generated visualizations (Event Cancellation, Festival Cover) and "Live Stats" bars.
- **Intelligent Insurance Calendar:** A custom-built visual calendar for customers that highlights upcoming premium payments, policy expiries, and critical notifications.
- **Risk-Based Premium Engine:** the system uses a 'Heuristics Function Logic - Compound Risk Multiplier' formula: $\text{Premium} = (\text{Coverage Amount} \times \text{Base Rate}) \times \Sigma(\text{Risk Factors})$.
- **Rich Detail Modals:** Deep-dive views for Claims and Policies, providing financial breakdowns (Settled vs Pending), active agent markers, and comprehensive audit logs.
- **Admin Analytics:** Global system health dashboards aggregating cross-table metrics, providing comprehensive operational insights.

UI/UX Excellence & Security

- **Aesthetics:** Modern "Glassmorphism" design, smooth CSS animations, and vibrant color palettes.
- **Responsive Design:** Fully optimized for Tablet, and Desktop.
- **Security Features:**
 - Masked passwords and visibility toggles.
 - MFA/Captcha/Bot Challenges on login.
 - Strict Role-Based Routing to prevent unauthorized data access.

Security & Stability Best Practices

Enterprise-grade protection.



- **Role-Based Access Control (RBAC):** All sensitive API endpoints are protected via `[Authorize(Roles = "...")]`, ensuring customers cannot access admin functions or claim officer logs.
- **Stateless JWT Authentication:** Passwords/tokens are never stored on the frontend; secure bearer tokens are used for every request with strict expiration and validation.
- **CORS (Cross-Origin Resource Sharing):** The API is strictly locked to only allow traffic from the authorized Angular frontend (`http://localhost:4200`).
- **Backend Testing:** Comprehensive xUnit + Moq unit test coverage, rigorously testing complex service logic directly for reliability.
- **Frontend Testing:** Behavior-Driven Development (BDD) testing utilizing standalone Jasmine frameworks, ensuring a seamless user experience.

Conclusion & Future Roadmap

Securing the Future of Events



Future Enhancements

Summary: EventSure successfully integrates complex insurance workflows into a single, intuitive platform.

- **Future Scope:**

- **Integration with real-time weather APIs for automatic cancellation triggers.**
- **Mobile App development using Ionic or Flutter.**
- **Automated Underwriting Logic:** Implementing AI-driven underwriting to minimize manual Agent assignments and accelerate policy issuance.
- **AI Chatbot Integration:** Deploying an intelligent chatbot for immediate public-facing customer support, enhancing accessibility and efficiency.

Final Quote: "Protecting your memories, one event at a time."

Thank You! Questions?

Thankyou!



The Hartford

Thank You

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